Eli's Math Review: Quiz 5

Name: _____ Date: _____

Simple Solutions Standards-Based Mathematics 4 — A Quiz #5 (Lessons 17–20). Every one of the 15 problems, with How-To pages, practice, a mini quiz, and an answer key at the back. ★ = extra practice (missed on the quiz).

Hi Eli!

You nailed rounding, the distributive property (1,384!), times-as-many, expanded form, the equilateral triangle, and drawing segment BC. Now let's make the tricky ones easy. Read each How-To box, do the practice, and always CHECK your work. You've got this!

What's in this packet

• Rounding • Expanded form	#1, #13	p. 2
★ Subtracting & adding big numbers (regrouping)	#2, #10 ★	p. 3
• Lines, segments, rays • Times as many	#3, #5	p. 4
• Distributive property (area model)	#4	p. 5
★ Perimeter: rectangles, squares, triangles	#6 ★, #15	p. 6
★ Equal fractions • Picking the right equation	#7, #8 ★	p. 7
★ Factor pairs (factor rainbow)	#9 ★	p. 8
★ Money: quarters • Inches, feet, yards	#11 ★, #12 ★	p. 9
★ Adding 4 numbers & estimating (more or fewer?)	#14 ★	p. 10
• Mini quiz (all 15 skills)	all	p. 11
• Answer key (for grown-ups)		p. 12

Grown-up note

Score 8/15. Marked wrong (red): #6 perimeter (only part of the way around; 22 cm), #8 picked B 15 + 9 = h and h = 24 (answer A, h = 6), #9 factor pairs of 24 (incomplete; 1&24, 2&12, 3&8, 4&6), #10 71,888 + 86,655 (carrying slip; 158,543), #11 quarters (23), #12 36 in = 1 yard (wrote 54), #14 more/fewer than 800 (more; 895). Also check #2: his answer looks like 1,087 (correct 1,086), but it was not marked. Interactive practice: 'Math review: Quiz 5' on Eli's page.

Rounding • Expanded Form

Name: _____ Date: _____

How-To: round to the nearest thousand (#1)



The halfway point is 5,500. 5,262 is closer to 5,000.

1. Which two thousands is it between? 5,000 and 6,000.
2. Look at the HUNDREDS digit: 5,262 → 2.
3. 0–4 round DOWN, 5–9 round UP.

5,262 → 5,000

Practice: round to the nearest thousand.

1. 7,815 → _____
2. 3,499 → _____
3. 9,620 → _____
4. 2,500 → _____
5. 6,051 → _____
6. 4,387 → _____

How-To: expanded form → standard number (#13)

6,000 + 400 + 20 + 7 → 6 thousands, 4 hundreds, 2 tens, 7 ones → 6,427

If a place is missing, write a 0 there: 9,000 + 50 + 3 = 9,053 (zero hundreds).

Practice: write the number.

7. 3,000 + 200 + 80 + 1 = _____
8. 9,000 + 50 + 3 = _____
9. 40,000 + 7,000 + 600 + 5 = _____
10. 8,000 + 900 + 4 = _____
11. 5,000 + 70 = _____
12. 20,000 + 3,000 + 10 + 9 = _____

Big Numbers: Regrouping

Name: _____ Date: _____

Subtract (#2)

$$\begin{array}{r} 3,072 \\ -\ 1,986 \\ \hline 1,086 \end{array}$$

Ones: 2 − 6? Borrow → 12 − 6 = 6. Tens: 6 − 8? Borrow → 16 − 8 = 8. Hundreds: 9 − 9 = 0. Thousands: 2 − 1 = 1.

Add (#10) ★

$$\begin{array}{r} {\scriptstyle 1\ \ 1\ 1} \\ 71,888 \\ +\ 86,655 \\ \hline 158,543 \end{array}$$

Line up the places. Add from the RIGHT: 8+5 = 13 → write 3, carry 1. Keep carrying! Last: 7 + 8 = 15 → write 15.

CHECK: subtract → add back (1,086 + 1,986 = 3,072). Add → estimate (70,000 + 90,000 = 160,000, so 158,543 makes sense).

Practice: subtract. Then check by adding.

1. $\begin{array}{r} 4,072 \\ -\ 1,986 \\ \hline \end{array}$

2. $\begin{array}{r} 5,003 \\ -\ 2,748 \\ \hline \end{array}$

3. $\begin{array}{r} 6,210 \\ -\ 3,475 \\ \hline \end{array}$

4. $\begin{array}{r} 8,000 \\ -\ 4,362 \\ \hline \end{array}$

Practice: add. Carry carefully! ★

5. $\begin{array}{r} 45,679 \\ +\ 38,547 \\ \hline \end{array}$

6. $\begin{array}{r} 62,918 \\ +\ 29,384 \\ \hline \end{array}$

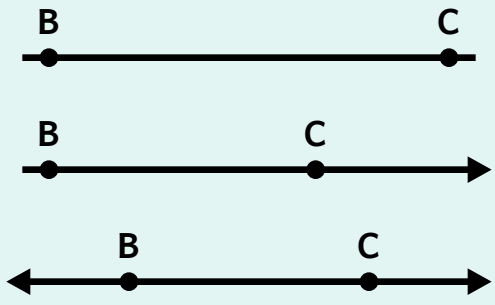
7. $\begin{array}{r} 71,888 \\ +\ 86,655 \\ \hline \end{array}$

8. $\begin{array}{r} 38,765 \\ +\ 47,456 \\ \hline \end{array}$

Lines, Segments, Rays • Times As Many

Name: _____ Date: _____

How-To: lines, segments and rays (#3)



Line segment BC 2 end points. Stops at both ends.

Ray BC 1 end point; goes on forever one way.

Line BC No end points. Arrows both ways.

Practice: draw each one.

1. Line segment XY
2. Ray PQ
3. Line MN

How-To: 'times as many' means MULTIPLY (#5)

●●●● ●●●● ●●●● ●●●● ●●●● ●●●● ●●●● **7 × 4 = 28**

7 groups of 4

Seven times as many as 4 → 7 × 4 = 28. It's NOT 7 + 4. Write the number sentence, then solve it.

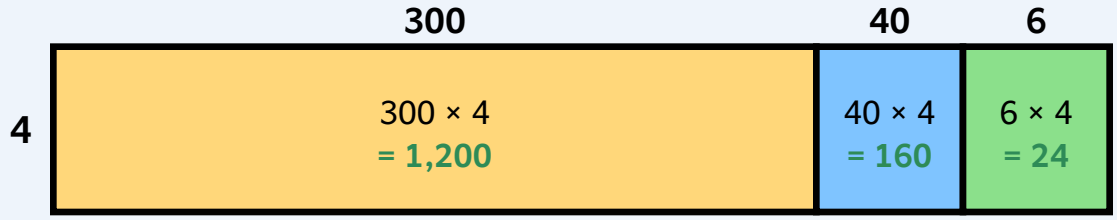
Practice: write a number sentence and solve.

4. Six times as many as 5 _____
5. Three times as many as 9 _____
6. Eli has 6 stickers. Owen has 3 times as many. How many does Owen have? _____
7. A dog weighs 8 times as much as a 4-pound cat. How much does the dog weigh? _____

Distributive Property: Break It Apart

Name: _____ Date: _____

How-To: 346 × 4 (#4)



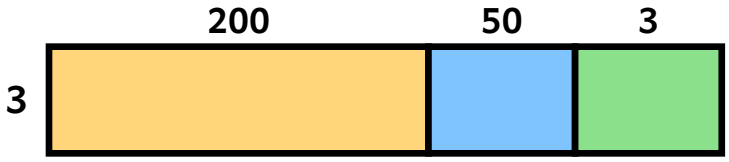
Break 346 into 300 + 40 + 6. Multiply EACH part by 4, then add the parts:

1,200 + 160 + 24 = 1,384

You got this one right on the quiz. Great job!

Practice: fill in the *area* model, then add.

1 **253 × 3 = (200 × 3) + (50 × 3) + (3 × 3)**



_____ + _____ + _____
= _____

2 **418 × 5 = (400 × 5) + (10 × 5) + (8 × 5)**



_____ + _____ + _____
= _____

3 **127 × 6 = (100 × 6) + (20 × 6) + (7 × 6)**



_____ + _____ + _____
= _____

4 **562 × 4 = (500 × 4) + (60 × 4) + (2 × 4)**

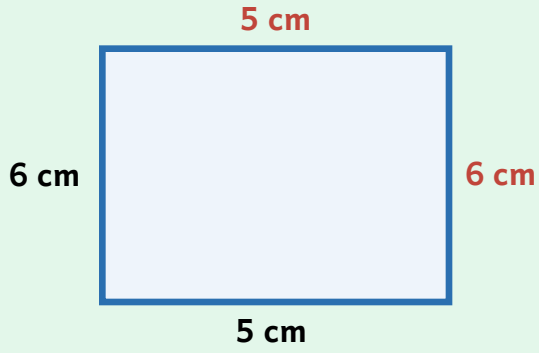


_____ + _____ + _____
= _____

Perimeter: All the Way Around ★

Name: _____ Date: _____

How-To: add EVERY side (#6, #15)



A rectangle has 4 sides. The quiz only labels 2 of them — the other two are the SAME (opposite sides match).

6 + 5 + 6 + 5 = 22 cm

Triangle with 3 equal sides of 12 m: 12 + 12 + 12 = 36 m.

Tip: trace around the shape with your finger and say each side.

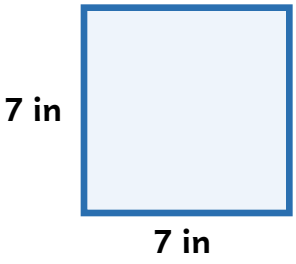
Practice: find the perimeter. Don't forget the unit!

1



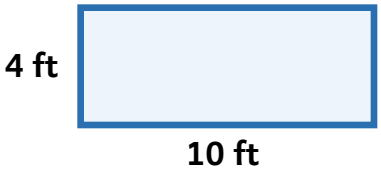
P = _____

2



P = _____

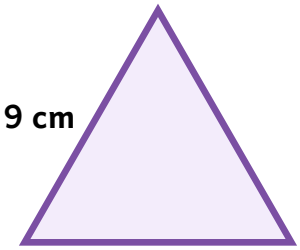
3



P = _____

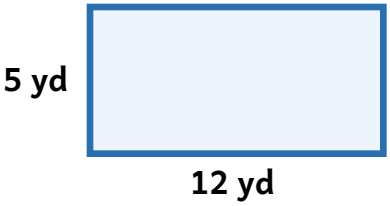
4

all sides equal



P = _____

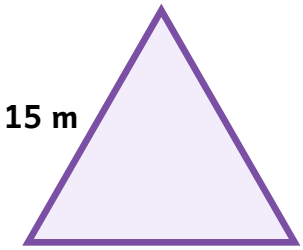
5



P = _____

6

all sides equal

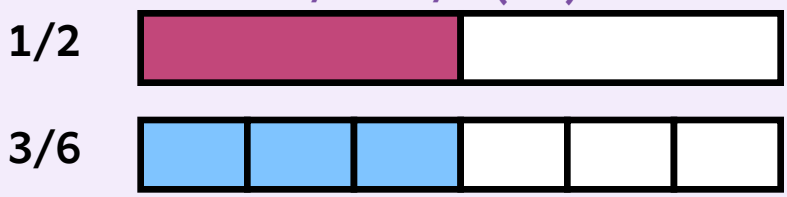


P = _____

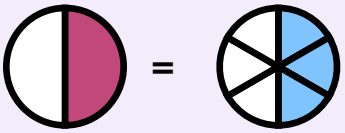
Equal Fractions • Pick the Right Equation ★

Name: _____ Date: _____

How-To: show 1/2 = 3/6 (#7)

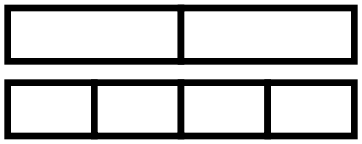


Same-size bars. Cut the half into 3 smaller pieces: 3 of 6 pieces = the SAME amount as 1 of 2.



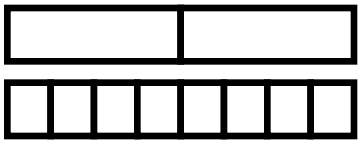
Practice: shade, then fill in the missing number.

1



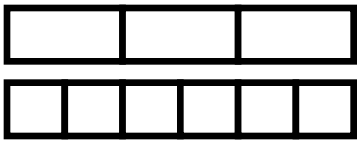
1/2 = ___/4

2



1/2 = ___/8

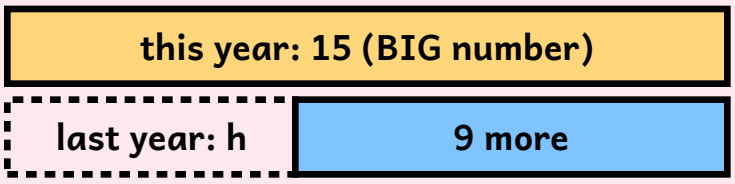
3



1/3 = ___/6

How-To: 'MORE than' problems (#8) ★

John hit 15 home runs. That is 9 MORE than last year. Last year = h.



h + 9 more = 15, so the equation is

A) 9 + h = 15

h = 6 (15 − 9). NOT 15 + 9 = 24!

Last year must be SMALLER than 15. If your answer is bigger than 15, check again.

Practice: circle the right equation, then solve.

4 Mia read 20 books. That's 8 more than Sam.

A) 8 + s = 20 B) 20 + 8 = s answer: _____

5 Max has 30 cards. That's 12 more than Lily.

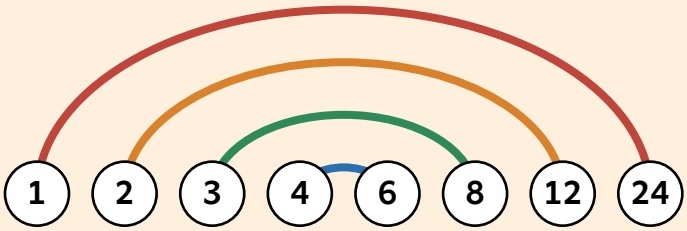
A) 30 + 12 = c B) 12 + c = 30 answer: _____

Factor Pairs: the Factor Rainbow ★

Name: _____ Date: _____

How-To: find ALL factor pairs of 24 (#9)

1. Start with 1: 1 × 24
2. Try 2: 2 × 12
3. Try 3: 3 × 8
4. Try 4: 4 × 6
5. Try 5: no (24 ÷ 5 doesn't work)
6. Next is 6 — already have it. STOP!



1 2 3 4 | 6 8 12 24 (8 factors, 4 pairs)

Write them in order: 1 & 24, 2 & 12, 3 & 8, 4 & 6. Don't stop after the first one!

Practice: list ALL the factor pairs. Then draw the rainbow.

1. **12:** ___×___ ___×___ ___×___

2. **18:** ___×___ ___×___ ___×___

3. **36:** ___×___ ___×___ ___×___ ___×___ ___×___

4. **20:** ___×___ ___×___ ___×___

5. **16:** ___×___ ___×___ ___×___

6. **30:** ___×___ ___×___ ___×___ ___×___

Quarters • Inches, Feet, Yards ★

Name: _____ Date: _____

How-To: quarters (#11) ★



4 quarters = $1.00

Step 1: How many quarters in $8.00? 8 × 4 = 32 quarters.

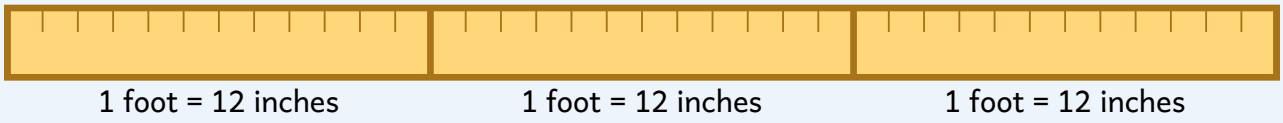
Step 2: She already HAS 9. How many MORE? 32 − 9 = 23 quarters.

Two-step problem: change dollars to quarters FIRST, then take away what she has.

Practice:

1. How many quarters make $5.00? _____
2. How many quarters make $10.00? _____
3. Ben needs $5.00 in quarters. He has 7. How many more does he need? _____
4. Zoe needs $3.00 in quarters. She has 4. How many more? _____

How-To: inches, feet, yards (#12) ★



12 inches = 1 foot 3 feet = 1 yard 1 yard = 36 inches

The cake is 36 inches tall = exactly 1 yard (one yardstick!).

Practice:

5. 2 yards = _____ inches
6. 72 inches = _____ yards
7. 1 yard = _____ feet
8. 4 feet = _____ inches
9. 6 feet = _____ yards
10. 3 yards = _____ feet

Adding Lots of Numbers: More or Fewer? ★

Name: _____ Date: _____

How-To: 347 + 220 + 148 + 180 — more or fewer than 800? (#14)

Way 1: Add exactly, two at a time

347	567	715
+ 220	+ 148	+ 180
567	715	895

895 > 800 → MORE

Way 2: Round to the nearest hundred to estimate

300 + 200 + 100 + 200 = 800. That is right AT 800, so add exactly!

When the estimate is right at the target, use Way 1. Answer the question in words: "They sold MORE than 800."

Practice: add exactly. Then answer MORE or FEWER.

1. 410 + 190 + 260 = _____ More or fewer than 900? _____

2. 235 + 318 + 156 + 94 = _____ More or fewer than 800? _____

3. 126 + 248 + 309 = _____ More or fewer than 700? _____

4. 480 + 215 + 199 = _____ More or fewer than 900? _____

Mini Quiz: Show What You Know!

Name: _____ Date: _____

1. Round 8,451 to the nearest thousand. _____
2. 6,040 − 2,875 = _____
3. Draw ray AB.
4. Use the distributive property: 237 × 3 = _____
5. Write and solve: five times as many as 6. _____
6. A rectangle is 9 ft by 4 ft. Perimeter = _____
7. Fill in: 1/2 = ____/10
8. Kim has 24 shells. That's 7 more than Jo. Which equation? A) 7 + j = 24 B) 24 + 7 = j. j = _____
9. List all the factor pairs of 20. _____
10. 56,749 + 37,586 = _____
11. Sam needs $6.00 in quarters. He has 11. How many more? _____
12. 48 inches = _____ feet; 2 yards = _____ inches
13. 7,000 + 300 + 8 = _____
14. 312 + 245 + 160 + 190: more or fewer than 900? _____
15. An equilateral triangle has sides of 14 cm. Perimeter = _____

Answer Key (grown-ups)

p.2 1) 8,000 2) 3,000 3) 10,000 4) 3,000 5) 6,000 6) 4,000 7) 3,281 8) 9,053 9) 47,605 10) 8,904 11) 5,070 12) 23,019

p.3 1) 2,086 2) 2,255 3) 2,735 4) 3,638 5) 84,226 6) 92,302 7) 158,543 8) 86,221

p.4 1–3) segment = 2 dots; ray = 1 dot + 1 arrow; line = 2 arrows. 4) 6 × 5 = 30 5) 3 × 9 = 27 6) 3 × 6 = 18 stickers 7) 8 × 4 = 32 pounds

p.5 1) 600 + 150 + 9 = 759 2) 2,000 + 50 + 40 = 2,090 3) 600 + 120 + 42 = 762 4) 2,000 + 240 + 8 = 2,248

p.6 1) 22 m 2) 28 in 3) 28 ft 4) 27 cm 5) 34 yd 6) 45 m

p.7 1) 2/4 2) 4/8 3) 2/6 4) A, s = 12 5) B, c = 18

p.8 1) 1×12, 2×6, 3×4 2) 1×18, 2×9, 3×6 3) 1×36, 2×18, 3×12, 4×9, 6×6 4) 1×20, 2×10, 4×5 5) 1×16, 2×8, 4×4 6) 1×30, 2×15, 3×10, 5×6

p.9 1) 20 2) 40 3) 20 − 7 = 13 4) 12 − 4 = 8 5) 72 6) 2 7) 3 8) 48 9) 2 10) 9

p.10 1) 860, fewer 2) 803, more 3) 683, fewer 4) 894, fewer

Mini quiz 1) 8,000 2) 3,165 3) dot A, through B, arrow past B 4) 600 + 90 + 21 = 711 5) 5 × 6 = 30 6) 26 ft 7) 5 8) A, j = 17 9) 1×20, 2×10, 4×5 10) 94,335 11) 24 − 11 = 13 12) 4 ft; 72 in 13) 7,308 14) 907, more 15) 42 cm

Original Quiz #5 answers

1) 5,000 2) 1,086 3) segment B•——•C 4) 1,200 + 160 + 24 = 1,384 5) 7 × 4 = 28 6) 22 cm 7) 1/2 shaded = 3/6 shaded

8) A: 9 + h = 15, h = 6 9) 1 & 24, 2 & 12, 3 & 8, 4 & 6 10) 158,543 11) 32 − 9 = 23 quarters 12) 1 yard

13) 6,427 14) 347 + 220 + 148 + 180 = 895 → more than 800 15) 12 + 12 + 12 = 36 m